

Provision Software with Terraform

To Begin with the Lab

Summary of the Lab

In this lab, we use Terraform to automatically create an AWS EC2 server and install Nginx on it without manually logging into the machine. A shell script is uploaded to the server and executed remotely, allowing Terraform to handle both infrastructure creation and software installation. The lab also covers creating SSH keys, configuring security group access, verifying that Nginx is running, and finally cleaning up all resources to avoid cloud charges.

- Create the required files:
 - provider.tf
 - create_instance.tf
 - variables.tf
 - install_nginx.sh
- Generate SSH Keys
- If keys do not exist, generate them.
- Command:

```
ssh-keygen -f levelup_key
```

- Files generated:
 - levelup_key
 - levelup_key.pub
- Initialize Terraform
- Command:

```
terraform init
```

- Review the Execution Plan
- Command:

```
terraform plan
```

- Create Infrastructure
- Command:

```
terraform apply
```

- Confirmation:

```
yes
```

- Verify Nginx Installation
- Open the EC2 public IP in a browser:
- <http://<EC2-Public-IP>>

- Expected result:
- Welcome to nginx!
- Remove All Resources
- Command:

terraform destroy

- Confirmation:

yes